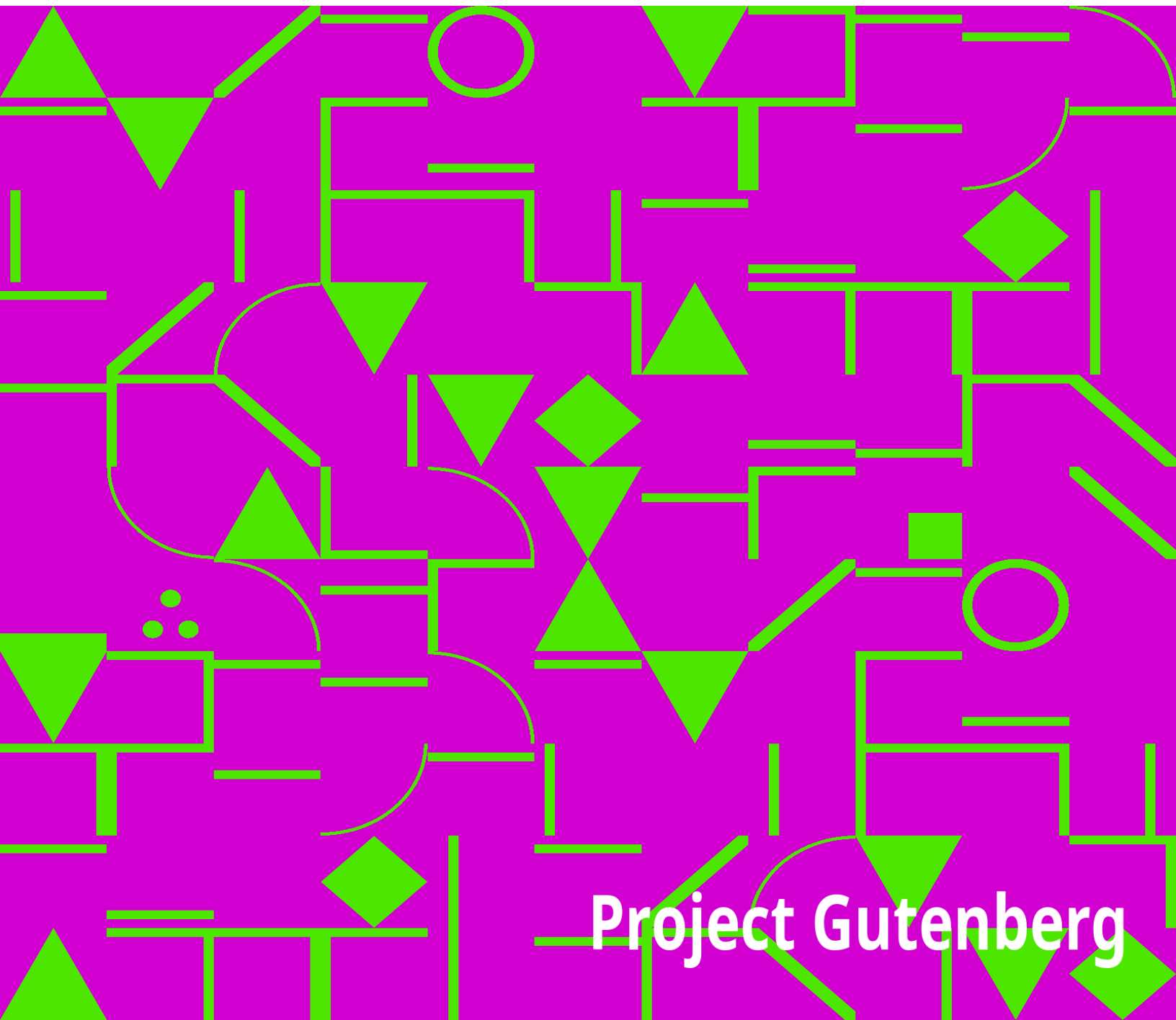


A New Species of Pocket Gopher (Genus *Pappogeomys*) From Jalisco, México

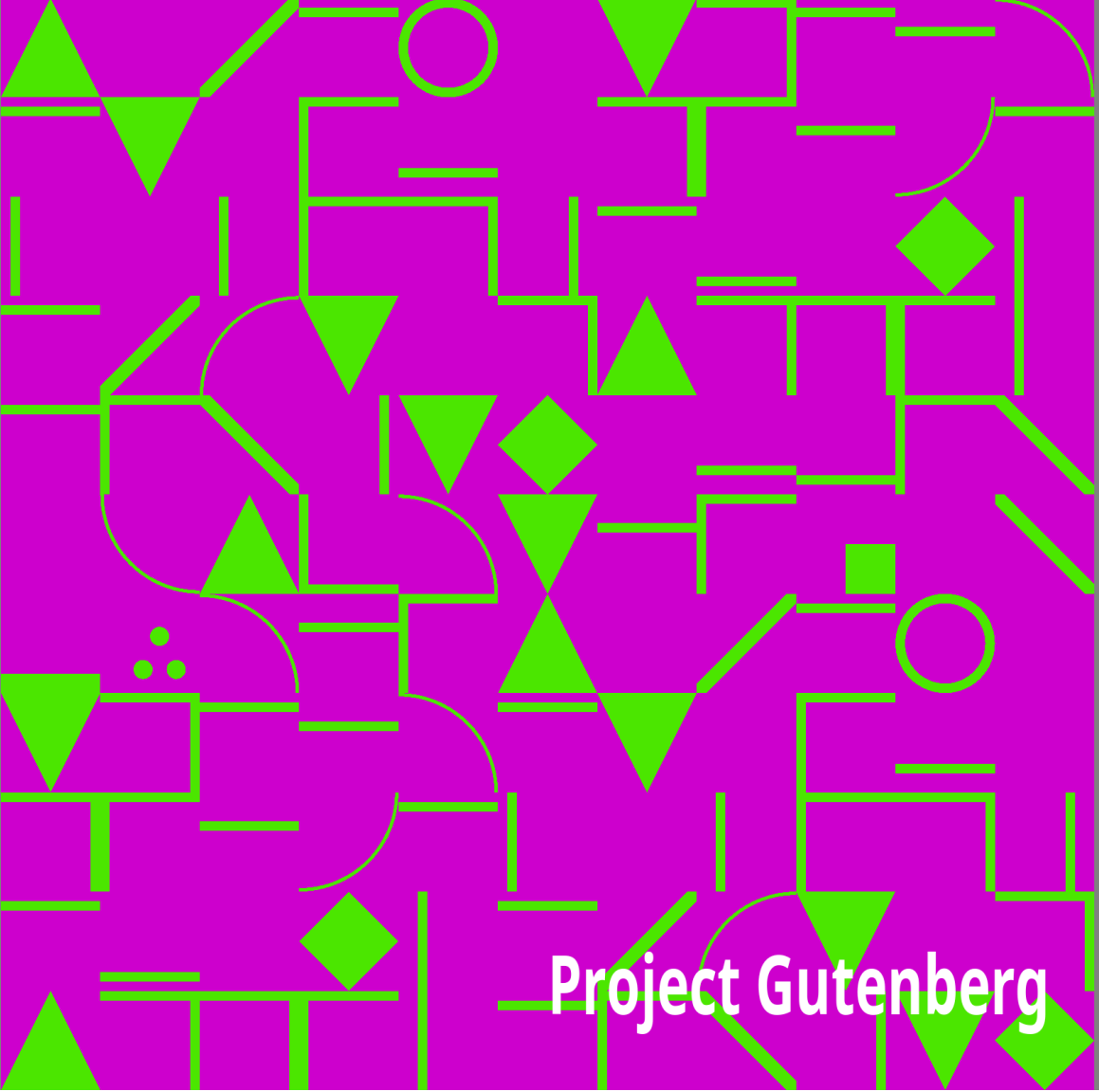
Robert J. Russell

An abstract geometric pattern consisting of various shapes like triangles, squares, circles, and lines in shades of blue, green, and yellow on a black background.

Project Gutenberg

A New Species of Pocket Gopher (Genus *Pappogeomys*) From Jalisco, México

Robert J. Russell

An abstract geometric pattern consisting of various shapes like triangles, squares, circles, and lines in shades of blue, green, and yellow, set against a black background.

Project Gutenberg

The Project Gutenberg eBook of A New Species of Pocket Gopher (Genus Pappogeomys) From Jalisco, México

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: A New Species of Pocket Gopher (Genus Pappogeomys) From Jalisco, México

Author: Robert J. Russell

Release date: January 20, 2010 [eBook #31025]
Most recently updated: January 6, 2021

Language: English

Other information and formats: www.gutenberg.org/ebooks/31025

Credits: Produced by Chris Curnow, Joseph Cooper, Diane Monico, and the Online Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK A NEW SPECIES
OF POCKET GOPHER (GENUS PAPPOGEOMYS) FROM JALISCO,
MÉXICO ***

UNIVERSITY OF KANSAS PUBLICATIONS
MUSEUM OF NATURAL HISTORY
Volume 9, No. 11, pp. 357-361
January 21, 1957

**A New Species of Pocket Gopher
(Genus
Pappogeomys) From Jalisco,
México**

BY
ROBERT J. RUSSELL

UNIVERSITY OF KANSAS
LAWRENCE
1957

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

**Editors: E. Raymond Hall, Chairman, Henry S. Fitch,
Harrison B. Tordoff**

**Volume 9, No. 11, pp. 357-361
Published January 21, 1957**

**UNIVERSITY OF KANSAS
Lawrence, Kansas**

**PRINTED BY
FERD VOILAND, JR., STATE PRINTER
TOPEKA, KANSAS
1957**

26-5801

A New Species of Pocket Gopher (Genus *Pappogeomys*) From Jalisco, México

BY

ROBERT J. RUSSELL

J. R. Alcorn collected a number of pocket gophers of the genus *Pappogeomys* in the states of Jalisco, Nayarit, and Colima. The bulk of this material was obtained in 1949 and 1950. Full treatment of these interesting pocket gophers will be given by the author in a future publication.

Among the *Pappogeomys* collected by Alcorn were three specimens from the high Sierra del Tigre, an isolated range not previously sampled for pocket gophers. The Sierra del Tigre is situated in southern Jalisco and western Michoacán, and, like most of the mountainous terrain in this region of México, is volcanic in origin. To the south the Sierra del Tigre descends abruptly to lower elevations of the arid coastal plains, uninhabited by gophers of this genus. The small pocket gopher occurring in the Sierra del Tigre seems to be an undescribed species of the genus *Pappogeomys* which may be known as

***Pappogeomys alcorni* new species**

Type.—Adult female, skull and skin; No. 39806, University of Kansas, Museum of Natural History; 4 mi. W Mazamitla, 6600 ft., Jalisco, México; October 18, 1950; obtained by J. R. Alcorn, original number 12835.

Distribution.—Known only from the Sierra del Tigre, and probably occurs only at higher elevations within the geographic limits of this isolated range of mountains.

Diagnosis.—Size medium for genus (see measurements); tail naked, short relative to length of head and body; hind foot short; hairs of upper parts and underparts Plumbeous basally and

Orange-Cinnamon apically (capitalized color terms after Ridgway, Color Standards and Color Nomenclature, Washington, D. C., 1912); large nasal patch Cinnamon-Buffer in two specimens, but Pale Pinkish-Buffer in holotype; white throat spot small and inconspicuous, throat mostly bright Cinnamon-Buffer; auricular patch pure Plumbeous, hairs lacking cinnamon-colored tips; tarsi with Cinnamon-Buffer hairs; dentition as in *P. bulleri* except that enamel plate of posterior wall of M1 reduced to a vestige present only on inner fourth, outer three-fourths of posterior wall of M1 without trace of enamel; zygomata slender, bowed outward; jugal long, widely separating maxillary and squamosal arms of zygoma; skull deep (measured from a point on the frontal to a point on the palate directly below and between the maxillary teeth); rostrum narrow and short; nasals broadly truncate posteriorly, and not decurved anteriorly; narrow across mastoid processes of squamosals; anterior palatine foramina small and rounded in outline, not slitlike.

Comparisons.—Compared with *Pappogeomys bulleri*, the only other named species of the genus, *P. alcorni* differs, as follows: Nasal patch cinnamon or buffy instead of white; enamel plate of posterior wall of M1 reduced to inner fourth rather than developed completely across posterior wall of tooth; nasals broadly truncate posteriorly instead of narrow and emarginate; anterior palatine foramina short and round instead of long and slitlike.

Measurements.—The type and an adult female topotype (in parentheses) measure, as follows: Total length, 210 (210); length of tail, 61 (63); length of hind foot, 29 (28); condylobasal length, 38.0 (36.9); basilar length, 32.8 (31.9); breadth across zygomata, 24.2 (24.8); palato-frontal depth, 15.0 (14.8); palatal length, 24.7 (24.1); length of nasals, 12.7 (12.8); breadth of braincase, 18.1 (17.5); breadth across mastoid processes of squamosals, 21.5 (21.4); breadth of rostrum, 8.4 (8.1); length of rostrum, 16.9 (16.3); alveolar length of maxillary tooth-row, 9.3 (8.8); breadth across angular processes of rami, 26.1 (26.2).

Specimens examined.—Three, all from Jalisco, as follows: 4 mi. W Mazamitla, 2; 3 mi. WSW Mazamitla, 1.

Remarks.—The features which distinguish *Pappogeomys alcorni* seem to be beyond the range of variation in *Pappogeomys bulleri*. In view of the absolute quality of the differences between *P. alcorni* and *P. bulleri*, it seems best to regard the former as a species, rather than as a subspecies of *P. bulleri*. Moreover, it seems unlikely that actual intergradation of the two species can occur, since the broad, low valleys between the higher terrain, where pocket gophers of this genus are found, do not offer suitable habitat for *Pappogeomys*.

In every example of *P. bulleri* that I have seen (more than 100 specimens, representing all of the named subspecies) the anterior palatine foramina are long and slitlike and the nasals are always narrow and emarginate posteriorly, whereas in *P. alcorni* the anterior palatine foramina are short and round and the nasals broad and squarely truncate posteriorly. The conspicuous nasal patch of *P. alcorni* is large and bright cinnamon or buffy, and, although the nasal patch may be large in some subspecies of *P. bulleri*, in each specimen possessing the patch the hairs are whitish with little or no trace of pigmentation.

One of the most interesting features of *P. alcorni* is the reduction of enamel on the posterior wall of the first upper molar. In *P. alcorni* the enamel present is thick, but it occurs only on the inner one-fourth of the posterior wall of the tooth. The enamel is always complete in *P. bulleri*; but in some old individuals it becomes thin with wear, and at a casual glance may appear to be partly or entirely absent. Close examination under magnification reveals, however, in every specimen of *P. bulleri*, a fine line of enamel completely across the posterior wall. It seems that the posterior enamel plate of M1 is disappearing in both *P. bulleri* and *P. alcorni*. In both species the enamel on the posterior wall of M1 does not extend down the crown so far as the level of the alveolus, whereas the anterior plate of enamel on M1, for example, extends well below the alveolus of the tooth. Even though disappearance of the posterior enamel seems to be a trend in both species, it has proceeded farther in *P. alcorni* than in *P. bulleri*. Examination of the posterior wall of M1 in *P. alcorni* disclosed only the

vestige of enamel on the inner side of the tooth, and no enamel, not even a thin plate, was present on the remainder of the posterior wall of the tooth.

The name *alcorni* is proposed as a token of appreciation to Joseph Raymond Alcorn, whose collecting has greatly enriched our knowledge of the mammals of México.

Transmitted August 30, 1956.

26-5801

*** END OF THE PROJECT GUTENBERG EBOOK A NEW SPECIES
OF POCKET GOPHER (GENUS PAPPOGEOMYS) FROM JALISCO,
MÉXICO ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.